

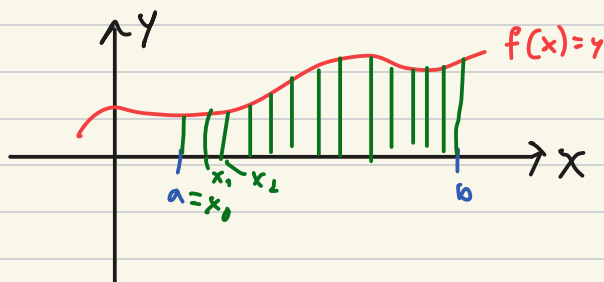
Integrals

From Calc I: Fundamental Theorem of Calculus:

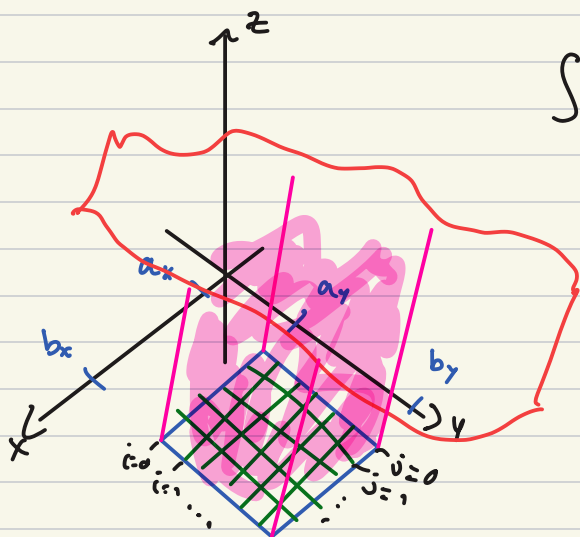
If $f: [a, b] \rightarrow \mathbb{R}$ is continuous then:

1) The function $F: [a, b] \rightarrow \mathbb{R}$ defined by $F(x) = \int_a^x f(t) dt$ is differentiable and $F'(x) = f(x)$ for all $x \in [a, b]$
(we say F is an antiderivative of f)

2) For any function F that is an antiderivative of f (i.e. $F'(x) = f(x)$) we have $\int_a^b f(x) dx = F(x) \Big|_{x=a}^{x=b} = F(b) - F(a)$

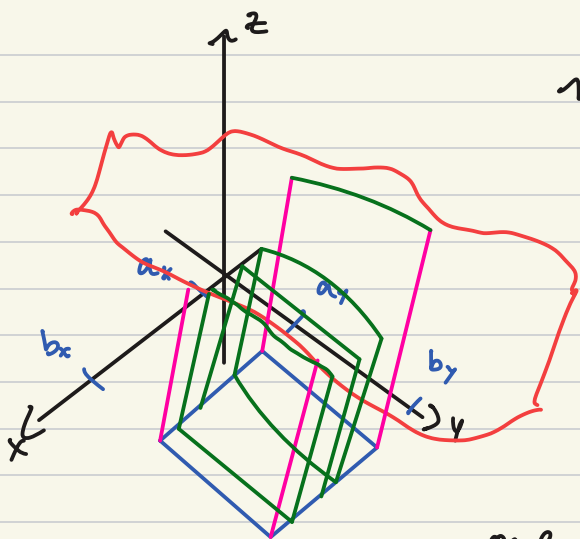


$$\int_a^b f dx = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(x_i) \Delta x_i$$



$$\int_{a_x}^{b_x} \left(\int_{a_y}^{b_y} f(x, y) dy \right) dx$$

$$\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \sum_{i,j=1}^n f(x_i) \Delta y_i \Delta x_j$$



or slices
slices: we cut the volume
by sections $x = c$

each slab has
area $\int_{a_y}^{b_y} f(x_i, y) dy$
and depth Δx_i

over a rectangular region

$$\{(x, y) : a_x \leq x \leq b_x, a_y \leq y \leq b_y\}$$

we can cut slabs perpendicular to x , or
perpendicular to y and
the integral (limit of Riemann sum) is
the same

e.g. $f(x, y) = 2x + y$ we want to find

the volume under this surface over
the rectangle $\{(x, y) : 2 \leq x \leq 4, 3 \leq y \leq 6\}$

write this as a double integral:

1) sections perpendicular to x

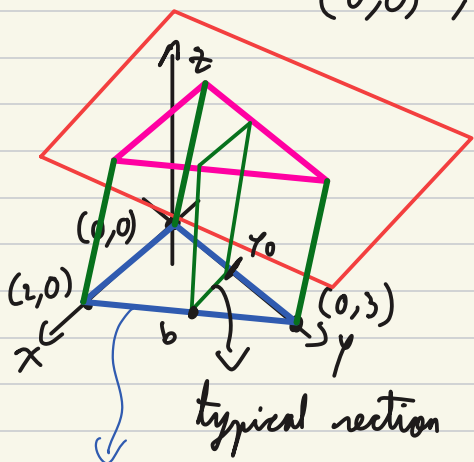
$$V = \int_2^4 \int_3^6 f(x, y) dy dx$$

2) sections perpendicular to y

$$\int_3^6 \int_2^4 f(x, y) dx dy$$

e.g. Volume of the region under $f(x,y)=1$

over the triangular region with vertices
 $(0,0)$, $(2,0)$, $(0,3)$



take slices perpendicular to y

typical section: x runs over $[0, b]$

this line has eqn. $y = 3 - \frac{3}{2}x$ or $b = 2 - \frac{2}{3}y$
 $x = 2 - \frac{2}{3}y$

$$\int_0^3 \int_0^{2-\frac{2}{3}y} 1 \, dx \, dy = \int_0^2 \int_0^{3-\frac{3}{2}x} 1 \, dy \, dx$$

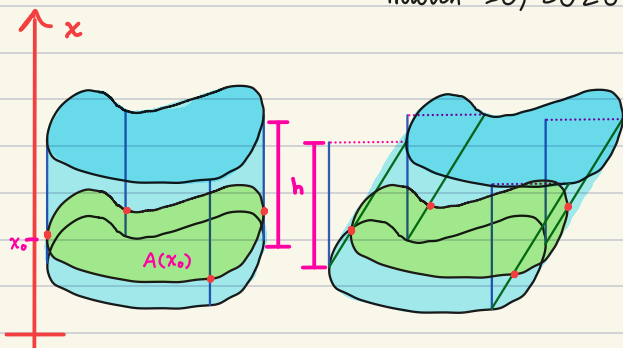
Integrals and Volumes

March 30, 2026

Cavalieri's Principle:

Two 3-D solids with the same height h and with the same cross sectional area (perpendicular to the height) have the same volume.

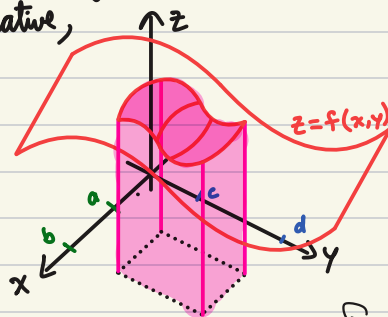
$$V = \int A(x) dx$$



Theorem: Suppose $\mathcal{R} = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$ and $f: \mathcal{R} \rightarrow \mathbb{R}$ is continuous and non-negative, then the volume of the region over the rectangle $\mathcal{R}$ and between the x - y plane and the surface $z = f(x, y)$ is

$$V = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

iterated integral



e.g.: Volume of the region bounded by the x - y plane and the surface $z = f(x, y) = y^3 - 9y^2 + 24y + 2x$ over the rectangle $2 \leq x \leq 4$ and $2 \leq y \leq 5$.

$$\begin{aligned} V &= \int_2^5 \int_2^4 (y^3 - 9y^2 + 24y + 2x) dx dy = \int_2^5 \left[\int_2^4 (y^3 - 9y^2 + 24y + 2x) dx \right] dy \\ &= \int_2^5 \left[(y^3 - 9y^2 + 24y)x + x^2 \right]_{x=2}^{x=4} dy = \int_2^5 \left[(y^3 - 9y^2 + 24y)4 + 16 \right] - \left[(y^3 - 9y^2 + 24y)2 + 4 \right] dy \\ &= \int_2^5 2(y^3 - 9y^2 + 24y) + 12 dy = \left. \frac{1}{2}y^4 - 6y^3 + 24y^2 + 12y \right|_{y=2}^{y=5} \\ &= \left[\frac{1}{2}(625) - 6(125) + 24(25) + 12(5) \right] - \left[\frac{1}{2}(16) - 6(8) + 24(4) + 12(2) \right] \\ &= \frac{1}{2}(609) - 6(117) + 24(21) + 12(3) = 304.5 - 702 + 504 + 36 = 1546.5 \end{aligned}$$

Rigorous definition of double integral and difference with iterated integrals

Recall the idea of area integral over $\overline{X} = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}$

or the limit of any Riemann sum of the form

$$\lim_{\substack{\Delta x_i \rightarrow 0 \\ \Delta y_i \rightarrow 0}} \sum_{i,j=0}^{m,n} f(x_i, y_j) \Delta x_i \Delta y_j \quad \begin{matrix} x_0 = a, x_m = b \\ y_0 = c, y_n = d \end{matrix}$$

This is the rigorous way of defining Riemann integral
in $\mathbb{R}^2 \leadsto$ denoted $\iint_{\overline{X}} f(x, y) dA$

we relate this to iterated integrals by taking "iterated limits".

$$\begin{aligned} & \lim_{\Delta x_i \rightarrow 0} \sum_{i,j=0}^{m,n} f(x_i, y_j) \Delta x_i \Delta y_j \\ &= \lim_{\Delta x_i \rightarrow 0} \sum_{i=0}^m \left(\sum_{j=0}^n f(x_i, y_j) \Delta y_j \right) \Delta x_i \\ &= \int_a^b \left(\sum_{j=0}^n f(x, y_j) \Delta y_j \right) dx \end{aligned}$$

$$\text{and } \lim_{\Delta y_i \rightarrow 0} \left(\lim_{\Delta x_i \rightarrow 0} \sum_{i,j=0}^{m,n} f(x_i, y_j) \Delta x_i \Delta y_j \right) = \int_a^b \int_c^d f(x, y) dy dx$$

as the T theorem in the previous page states,

this can be done for any f which is a continuous function.

intuitively $\Delta x_i \rightarrow 0$ produces dx (length element along x)
 $\Delta y_i \rightarrow 0$ produces dy (length element along y)
and $\Delta x_i \rightarrow 0$ taken to $dA = dx dy$ (area element in $\mathbb{R}^2$)
 $\Delta y_i \rightarrow 0$

Properties of Riemann integral in $\mathbb{R}^2$

Theorem: Suppose f and g are Riemann integrable on a region $R \subseteq \mathbb{R}^2$, Then

1) $f + g$ is also integrable and

$$\iint_R (f+g) dA = \iint_R f dA + \iint_R g dA$$

2) For any $k \in \mathbb{R}$, $k \cdot f$ is also integrable and

$$\iint_R k f dA = k \iint_R f dA$$

3) If $f(x,y) \leq g(x,y) \quad \forall (x,y) \in R$, then

$$\iint_R f dA \leq \iint_R g dA$$

4) If there is $M > 0$ so that $|f(x,y)| \leq M$

for all $(x,y) \in R$, then $|\iint_R f dA| \leq M \iint_R dA$

$$\text{and } |\iint_R f dA| \leq \iint_R |f| dA.$$

5) If R_1 and R_2 are disjoint regions in $\mathbb{R}^2$ (i.e. $R_1 \cap R_2 = \emptyset$), and if f is integrable on each of R_1 and R_2 , then

$$\iint_{R_1 \cup R_2} f dA = \iint_{R_1} f dA + \iint_{R_2} f dA$$

(e.g. in $\mathbb{R}^1$ $\leftarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$)

e.g. Compute $\iint_R x^2 + y \, dA$ for $R = [-1, 2] \times [1, 4]$
" $\{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 2, 1 \leq y \leq 4\}$

$$\begin{aligned} \iint \dots dx dy &= \int_1^4 \left[\int_{-1}^2 (x^2 + y) dx \right] dy = \int_1^4 \left[\frac{x^3}{3} + yx \Big|_{x=-1}^{x=2} \right] dy \\ &= \int_1^4 \left(\frac{8}{3} + 2y \right) - \left(\frac{-1}{3} - y \right) dy = \int_1^4 3 + 3y dy \\ &= \left[3y + \frac{3}{2} y^2 \Big|_{y=1}^{y=4} \right] = 12 + 24 - 3 - \frac{3}{2} \\ &= 31.5 \end{aligned}$$

$$\iint \dots dy dx = \int_{-1}^2 \left[\int_1^4 (x^2 + y) dy \right] dx = \int_{-1}^2 \left[x^2 y + \frac{y^2}{2} \right]$$

T.B.C.